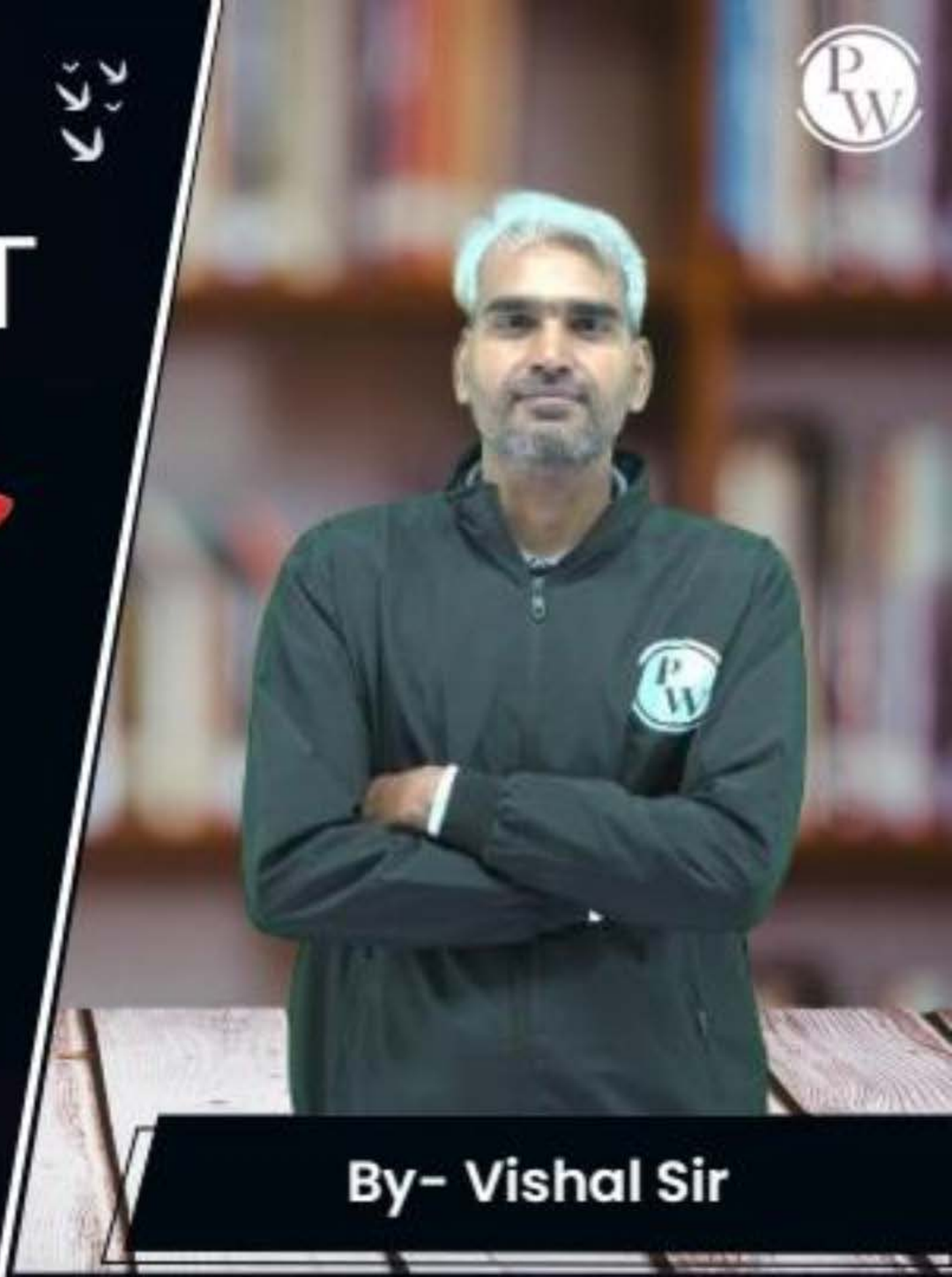
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 14



By- Vishal Sir

Recap of Previous Lecture



Topic Decomposition of relation



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Topics to be Covered

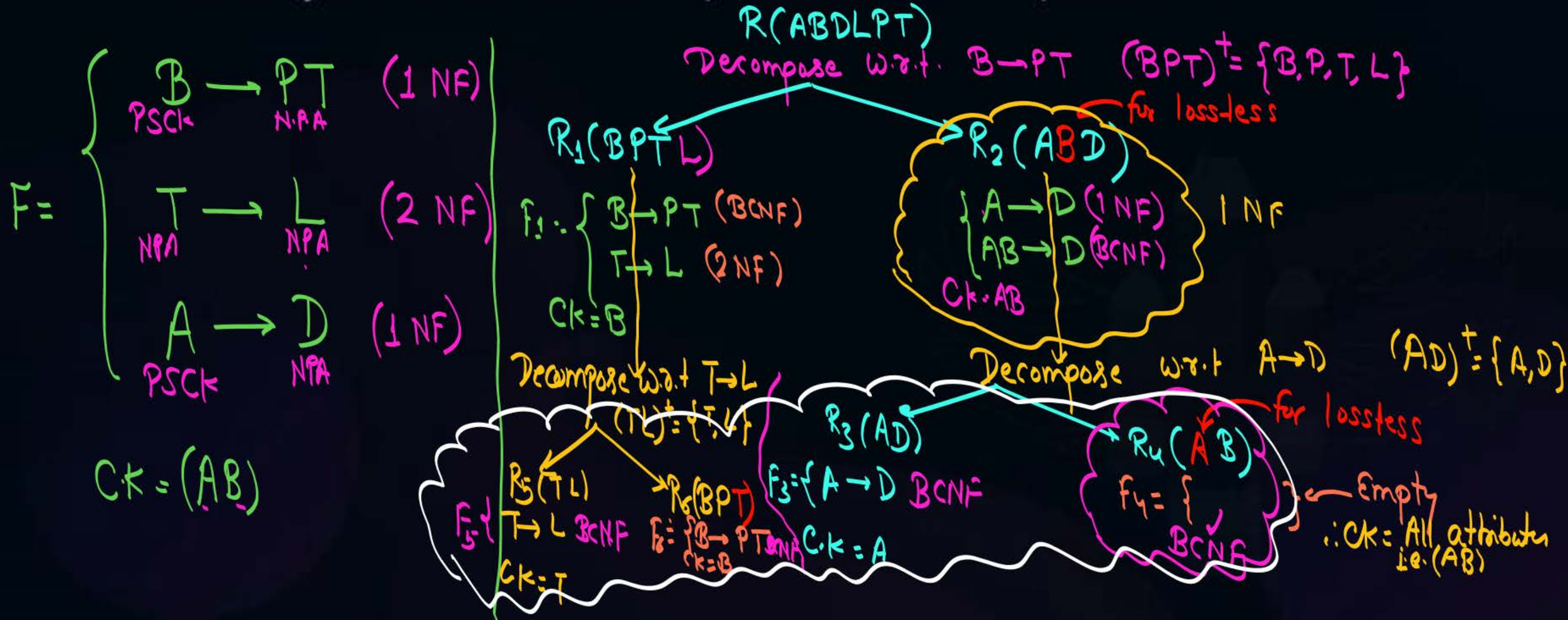


- Topic: Decomposition of relation
- Topic: Multivalued dependency
- Topic: Properties of multivalued dependency
- Topic: Fourth normal form (4NF)

Slide

#e.g. Given R(ABDLPT) and F={B→PT, T→L, A→D}

Find the highest normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.



Note:- If FD set of a relation does not contain any non-trivial FD then,

Candidate key of that relation will contain all attributes of that relation

and

Such relation will always be in BCNF (at least)

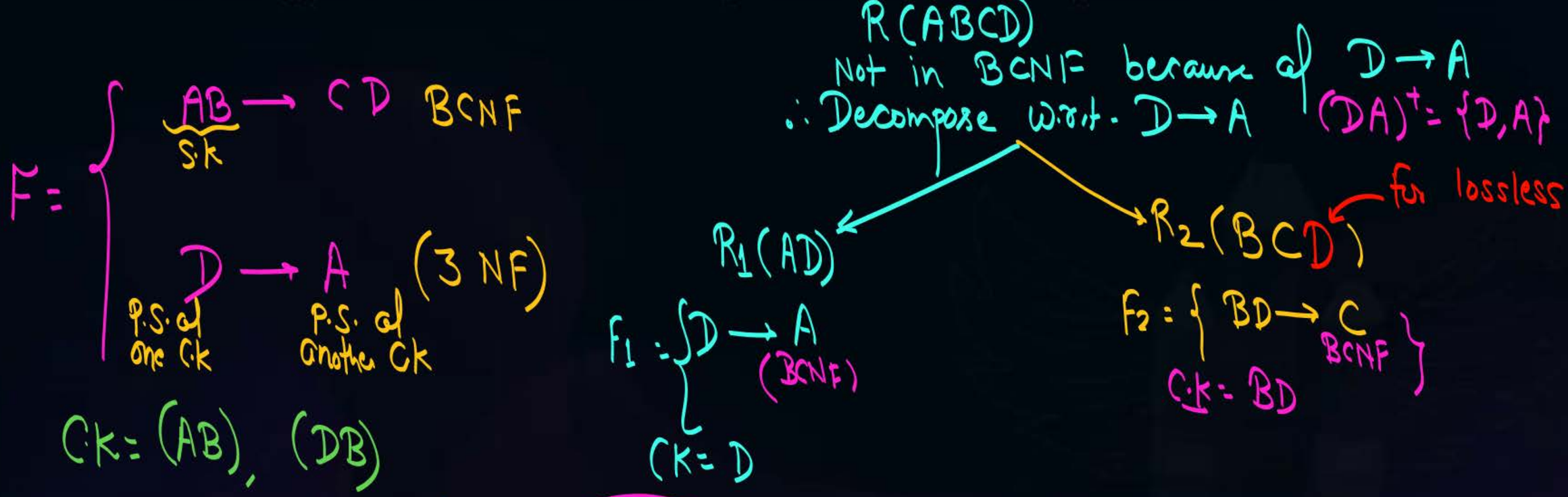
#e.g. Given R(ABCDE) and F:{A→BC, CD→E, B→D, E→A}

Find the highest normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

Recall, we had to create a new relation R'(CDE) to preserve FD CD → E

#e.g. Given R(ABCD) and F={AB→CD, D→A }

Find the highest normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.



If we tour to Preserve AB→CD by creating a new Rel^n R'(ABCD) then it will be in 3NF bcoz of D→A

(AB)+ w.r.t F1 ∪ F2 = {A, B} i.e. AB → CD is lost



Topic : NOTE

(1) We can always ensure Lossless Join decomposition while decomposing a relation into BCNF, but while trying to decompose a relation into BCNF sometimes we may loose some of the dependencies

(2) Upto 3NF we can ensure both Lossless Join decomposition as well as dependency preserving decomposition,
∴ Most adequate normal form of database is 3NF.



Topic : NOTE

③ If all attributes of a relation are prime attributes. then that relation is at least in 3 NF.

④ If all Candidate keys are simple Candidate keys then relation is at least in 2NF {because partial dep. is not possible}

⑤ If all attributes are prime attributes, and all Candidate keys are simple Candidate keys then relation is always in BCNF (at least)



Topic : NOTE

- A relation with only two attributes is called a binary relation

⑥ A binary relation is always in BCNF

eg: R(AB) is a binary relation 4 different FD sets are possible over rel^n R(AB)

F1 $\{A \to B$ BCNF, $B \to A$ BCNF$\}$

CK := (A), (B)

F2 $\{A \to B$ SK, BCNF$\}$

CK := (A)

F3 $\{B \to A$ SK, BCNF$\}$

CK = B

F4 { } empty

∴ BCNF

* If a relation is in BCNF then there will be no redundancy (0% redundancy) in the relation because of function dependencies,

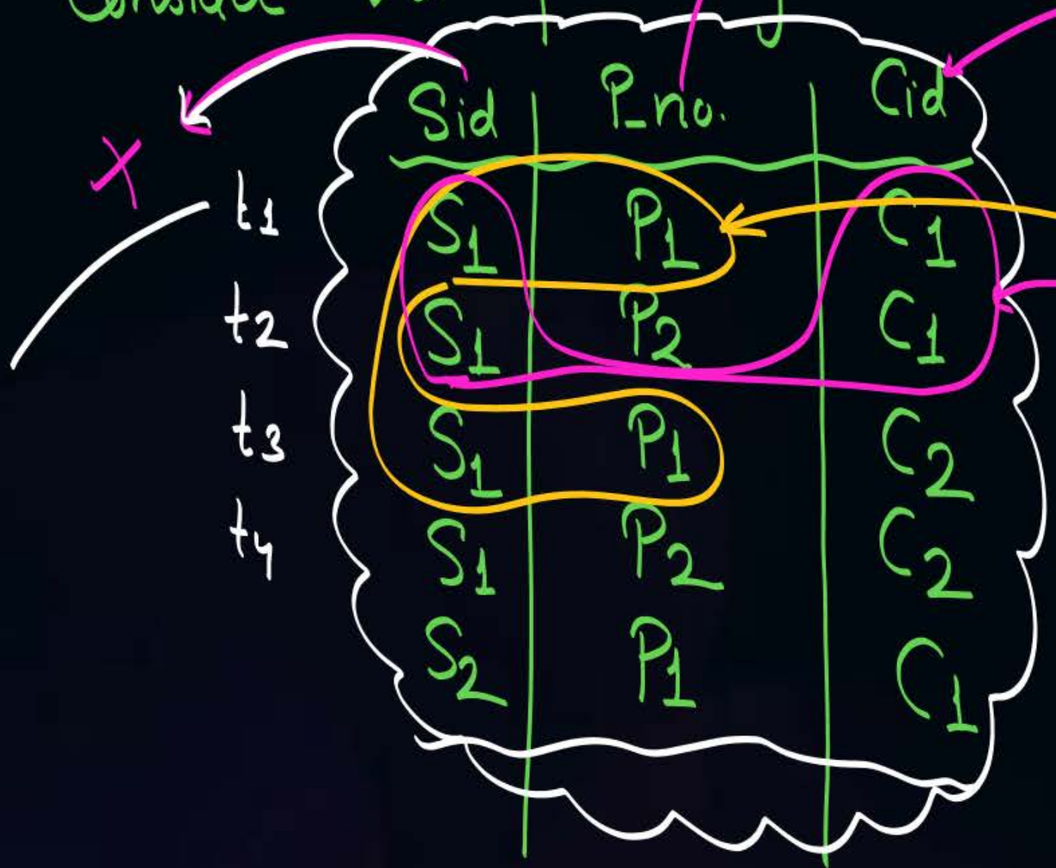
but such relations may have redundancy because of multi-valued dependency.



Topic : Multivalued dependency

X Multivalued determines Y is denoted as $X \twoheadrightarrow Y$

Consider the following relation



There is no non-trivial Functional dependency in the relation

∴ Relation is in BCNF

But redundancy is still present in the relation because of multi-valued dependency

If there exists two or more independent attributes which depends on some other attribute of the relⁿ then multivalued dependency may be present in the relation



Topic : Multivalued dependency

R(A B C D E F G H)
X
Y

Formal definition :- Let R be the relational schema and, X and Y be the attribute sets over relation R then we define Z = Attributes of R − (X ∪ Y)

- If there exists four tuples $t_1, t_2, t_3, t_4 \in R$ s.t.

(i) $t_1.X = t_2.X = t_3.X = t_4.X$

and (ii) $t_1.Y = t_3.Y$ & $t_2.Y = t_4.Y$

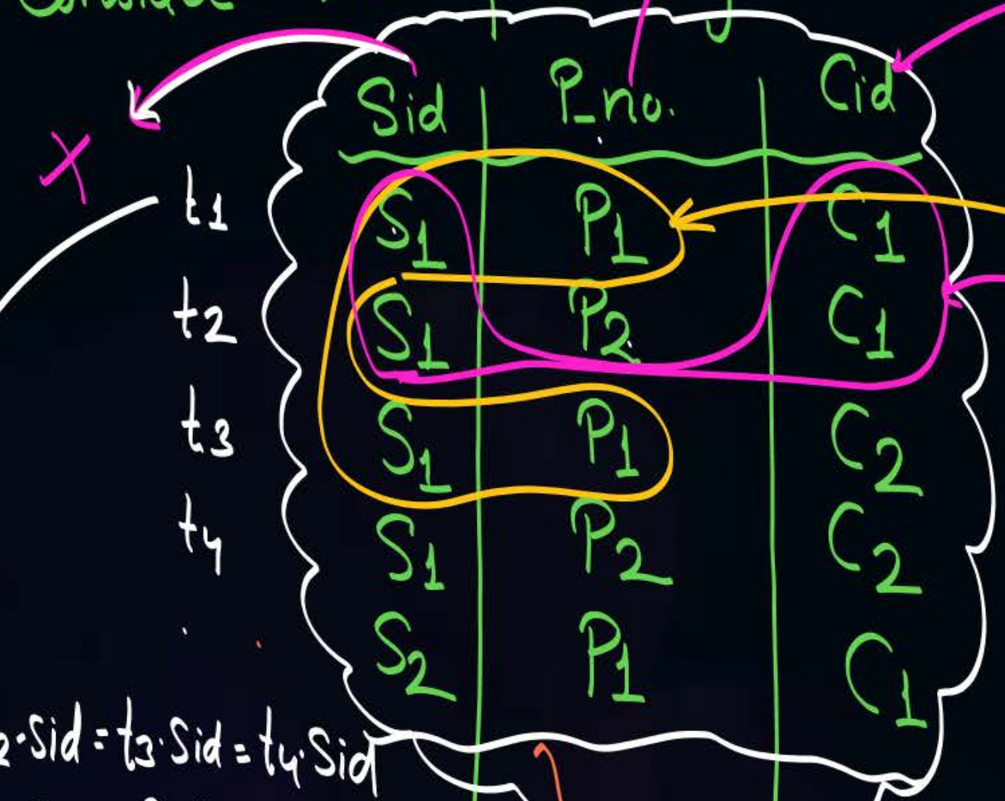
and (iii) $t_1.Z = t_2.Z$ & $t_3.Z = t_4.Z$

then we say tha X mutivalued determines Y.

i.e Multi-valued dependency "$X \twoheadrightarrow Y$" exists in R

Topic : Multivalued dependency

Consider the following relation



X Y Z

$t_1.Sid = t_2.Sid = t_3.Sid = t_4.Sid$

& $t_1.Pno = t_3.Pno$ & $t_2.Pno = t_4.Pno$

& $t_1.Cid = t_2.Cid$ & $t_3.Cid = t_4.Cid$

$\Rightarrow$ i.e. $Sid \rightarrow\!\!\rightarrow Pno$

There is no non-trivial Functional dependency in the relation

$\therefore$ Relation is in BCNF

But redundancy is still present in the relation because of multi-valued dependency



Topic : Multivalued dependency

Consider the following relation

Z

X

Y

	Sid	P_no	Cid
t_1	S_1	P_1	C_1
t_3	S_1	P_2	C_1
t_2	S_1	P_1	C_2
t_4	S_1	P_2	C_2
	S_2	P_1	C_1

$t_1.Sid = t_2.Sid = t_3.Sid = t_4.Sid$

& $t_1.Cid = t_3.Cid$ & $t_2.Cid = t_4.Cid$

& $t_1.P\text{-}no = t_2.P\text{-}no$ & $t_3.P\text{-}no = t_4.P\text{-}no$.

$\Rightarrow$ i.e. $Sid \twoheadrightarrow Cid$

There is no non-trivial Functional dependency in the relation

$\therefore$ Relation is in BCNF

But redundancy is still present in the relation because of multi-valued dependency



Topic : Multivalued dependency

Note: If We swap the values of attribute 'Y' in two tuples which agree of attribute X {i.e. tuple that have same value of X}, and if it does not create any new tuple in relation then we say

$$X \rightarrow\rightarrow Y$$



Topic : Properties of Multivalued dependency

① Complementation Rule :- let R be the relational schema and, X & Y be the attribute sets over relation R and $Z = R-(X \cup Y)$

If $X \twoheadrightarrow Y$, then $X \twoheadrightarrow Z$



Topic : Properties of Multivalued dependency

② Trivial MVD :-

$X \twoheadrightarrow Y$ is called a trivial MVD

if, $X \supseteq Y$ or XUY = overall attributes of Relⁿ

XUY covers all attributes of the Relⁿ



Topic : Properties of Multivalued dependency

③ Transitivity :

if $X \rightarrow\rightarrow Y$ and $Y \rightarrow\rightarrow Z$

then $X \rightarrow\rightarrow Z - Y$



Topic : Properties of Multivalued dependency

④ Augmentation:-

Let $X \rightarrow\rightarrow Y$ and $\alpha \supseteq \beta$

then $X\alpha \rightarrow\rightarrow Y\beta$



Topic : Properties of Multivalued dependency

(5) Replication :-

Every functional dependency $X \rightarrow Y$ is a multivalued dependency $X \twoheadrightarrow Y$, but

Every Multivalued dependency $X \twoheadrightarrow Y$ need not be FD $X \rightarrow Y$



Topic : Properties of Multivalued dependency

⑥ Splitting Rule:-

MVDs are not allowed to split.

We know

if $X \rightarrow YZ$ {FD

then $X \rightarrow Y$ & $X \rightarrow Z$ } holds true

Splitting is Valid w.r.t. FD.

If $X \rightarrow\rightarrow YZ$

then $X \rightarrow\rightarrow Y$ & $X \rightarrow\rightarrow Z$ need not hold true

Splitting is not allowed w.r.t. MVD



Topic : Fourth normal form (4NF)

A relation R is in 4NF only

(i) R must be in BCNF

(ii) Every non-trivial MVD $X \rightarrow\rightarrow Y$ must be with 'X' as a Super key

Super key is always calculated using functional dependencies, Because functional dependencies are "key" dependencies whereas MVDs are data dependencies.



Topic : Fourth normal form (4NF)

Sid	P_no.	Cid
S_1	P_1	C_1
S_1	P_2	C_1
S_1	P_1	C_2
S_1	P_2	C_2
S_2	P_1	C_1

No non-trivial FD
∴ Relation is in BCNF

Non-trivial MVDs in relation are

Sid →→ P_no. { neither trivial MVD nor Sid is S.K ⤷ ∴ Not in 4NF

Sid →→ Cid
Not a S.K ∴ Not in 4NF

Decompose w.r.t. Sid →→ P_no.

Decompose w.r.t. Sid →→ Cid

Sid	P_no
S_1	P_1
S_1	P_2
S_2	P_1

CK :-(Sid, P_no)

Sid →→ P_no still exists in the relation but now it is trivial (Sid ∪ P_no) = All attributes
↓
∴ 4NF

Sid	Cid
S_1	C_1
S_1	C_2
S_2	C_1

Sid →→ Cid but it is trivial
↓
∴ 4NF



2 mins Summary

Topic Decomposition of relation

Topic Multivalued dependency

Topic Properties of multivalued dependency

Topic Fourth normal form (4NF)

THANK - YOU



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